

Module 5 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 5, Quiz A: Definitions & notation

Reading: [Module 5 Read 1](#)

Project: [Project 5](#) · **Canvas:** Def & Notation quiz (Module 5)

Work these **practice** items after [Read 1](#). Sections follow Read 1: scatterplot vocabulary, regression line notation (slope, intercept, prediction), and residuals / MSE / R^2 .

Inference for slope, CI vs PI → [Quiz B](#) after [Read 2](#).

Section A — Scatterplot vocabulary (Read 1 §1)

A1. Form

Define **form** in the context of describing a scatterplot.

A2. Direction

Define **direction** in the context of a scatterplot.

A3. Strength

Define **strength** in the context of a scatterplot.

A4. Pearson correlation

Define **Pearson correlation** r and state its range.

A5. Positive vs negative correlation

What does the **sign** of r tell you? What does $|r|$ close to 1 vs close to 0 tell you?

Answer key — Section A — Scatterplot vocabulary (Read 1 §1)

1. **Form** describes the overall shape of the pattern — whether the relationship looks roughly **linear**, **curved**, or **neither** (no pattern).
 2. **Direction** is **positive** (both variables tend to increase together), **negative** (one increases as the other decreases), or **none** (no consistent direction).
 3. **Strength** describes how closely the points cluster around a trend — **strong** (tight cluster), **moderate**, **weak**, or **none**.
 4. Pearson r is a **unitless** measure of the **strength and direction** of a **linear** association between two numerical variables; it ranges from -1 to 1 .
 5. $r > 0$: positive direction (as x increases, y tends to increase). $r < 0$: negative. $|r|$ near 1 : strong linear association. $|r|$ near 0 : weak to no linear association.
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Section B — Regression line notation (Read 1 §1)

B1. Fitted line notation

Write the simple linear regression prediction equation and identify each symbol.

B2. Slope interpretation

State the general interpretation template for a slope b_1 in context.

B3. Intercept interpretation

State the general interpretation template for the intercept b_0 .

B4. Point prediction

Exam-score model: $\hat{y} = 47.7 + 3.06x$ (study hours). Predict the score for a student studying 8 hours.

B5. Fitting with `lm()`

Write the R call to fit a simple linear regression of `height_cm` on `dose` using a data frame `plant_dat`.

Answer key — Section B — Regression line notation (Read 1 §1)

1. $\hat{y} = b_0 + b_1x$. $\hat{y}$: predicted (fitted) value of the response. b_0 : **intercept** (predicted y when $x = 0$). b_1 : **slope** (predicted change in y per one-unit increase in x). x : value of the explanatory variable.
 2. For each additional one-unit increase in x , the predicted value of y changes by b_1 units (increases if $b_1 > 0$, decreases if $b_1 < 0$).
 3. The intercept b_0 is the predicted value of y when $x = 0$. Caution: this is only meaningful if $x = 0$ is a plausible value in context.
 4. $\hat{y} = 47.7 + 3.06(8) = 47.7 + 24.48 = 72.18$ points.
 5. `lm(height_cm ~ dose, data = plant_dat)` (or assign to an object: `fit <- lm(height_cm ~ dose, data = plant_dat)`).
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Section C — Residuals, MSE, and R^2 (Read 1 §1)

C1. Residual definition

Define a **residual** for an observation (x_i, y_i) with fitted value $\hat{y}_i$.

C2. Least-squares criterion

What does the least-squares method minimize to choose the best regression line?

C3. Test MSE

Define **test MSE** and explain why it is preferred over training MSE for evaluating a model.

C4. R^2 interpretation

Interpret $R^2 = 0.68$ on the test set for a study-hours $\rightarrow$ exam-score model.

Answer key — Section C — Residuals, MSE, and R^2 (Read 1 §1)

1. Residual $= y_i - \hat{y}_i$ — the difference between the **observed** response and the model's **predicted** response.
 2. Least-squares minimizes the **sum of squared residuals** $\sum (y_i - \hat{y}_i)^2$ — no other line produces a smaller total.
 3. Test MSE $= \frac{1}{n_{\text{test}}} \sum (y_i - \hat{y}_i)^2$ on held-out data. It is preferred because **training MSE is optimistically biased** — the model was fit on those same points, so training MSE understates error on new data.
 4. The linear model explains about **68%** of the variability in exam scores in the test set. The remaining 32% is unexplained by study hours alone.
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